Hotel Management System - Technology Stack

An overview of the core technologies, libraries, and tools used to build a robust, scalable, and modern Hotel Management application.

Core Framework & Build Tool

- **React (v19)**: The foundational library for building the user interface. Enables encapsulated, reusable UI components that manage their own state.
- **Vite**: A modern, blazing-fast frontend build tool that provides a significantly faster and leaner development experience compared to traditional bundlers.

Routing & Navigation

- **React Router DOM**: Enables dynamic routing within our Single-Page Application (SPA), allowing smooth transitions between views (like Dashboard, Rooms, Bookings) without page reloads.

Styling & UI Architecture

- **Tailwind CSS**: A utility-first CSS framework that speeds up UI development by providing low-level styling classes directly in the markup.
- **Lucide React**: A beautiful, consistent, and highly customizable icon library designed specifically for modern React applications.
- **clsx & tailwind-merge**: Utilities for conditionally joining CSS class names, ensuring dynamically applied Tailwind classes don't conflict.

State Management & Forms

- **React Hook Form**: A highly performant and flexible library for managing form state and validation, critical for complex input scenarios like booking forms.

Animations & Analytics

- **Framer Motion**: A powerful animation library for React to add complex, fluid, and interactive animations to components.
- **Recharts**: A composable charting library built on React, used for rendering data visualizations and analytics (e.g., occupancy rates).

Utility Libraries

- **date-fns**: A comprehensive and lightweight toolset for manipulating and formatting dates, essential for the booking schedule system.
- **Oxlint**: An extremely fast JavaScript linter written in Rust, used to catch errors and enforce code quality standards efficiently.

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